

UDDIPA DATTUDU CHAVALI

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ACADEMIC & RESEARCH PROFILE

Electronics and Communication Engineering student focusing on the VLSI domain. Gaining foundational knowledge in RFIC design and analog-to-digital converters through academic research internships. Capable of building, simulating, and designing custom layouts for hardware using Cadence Virtuoso and Verilog.

ACADEMIC BACKGROUND

Madanapalle Institute of Technology and Science

Bachelor of Technology in Electronics and Communication Engineering; CGPA: 9.37

Andhra Pradesh, India

July 2025 – Expected May 2028

- **Relevant Coursework:** Analog Circuits, Digital System Design (Verilog HDL), Electronic Devices & Circuits, Network Theory, Python Programming, PCB Design.

ESC Government Polytechnic College

Diploma in Electronics and Communication Engineering; Percentage: 95.2%

Andhra Pradesh, India

Aug 2022 – May 2025

TECHNICAL SKILLS

Core Competencies: Analog Circuit Design, Radio Frequency CMOS Integrated Circuit Design, DC/AC/Transient Simulations.

EDA Tools: Cadence Virtuoso, Altium Designer (Basic PCB), NGSPICE, NI Multisim, MATLAB.

Programming & HDLs: Verilog HDL, Python (Currently learning Data Science & Basic Automation), C language.

ACADEMIC, RESEARCH AND INDUSTRIAL EXPERIENCE

National Institute of Technology (NIT) Calicut

Research Intern - Analog IC Design for Image Sensors

Remote

July 2026 – Present

- Collaborating on remote research under the supervision of Dr. Bhuvan B, focusing on the study and design of Programmable Gain Amplifiers (PGAs) and Single-Slope ADCs for CMOS camera image sensors.
- Working on the transistor-level modeling and simulation of the converter blocks to optimize gain, noise, and power for high-resolution imaging applications.

National Institute of Technology (NIT) Calicut

Summer Research Intern - RFIC Design for Anti-Drone Radar

Kerala, India

May 2026 – July 2026

- Designed and simulated an unconditionally stable 9.0 GHz cascode Low Noise Amplifier (LNA) in a 180nm CMOS process using Cadence Virtuoso under the guidance of Dr. Karthik R, achieving a forward voltage gain (S_{21}) of 11.06 dB and a Noise Figure (NF) of 1.41 dB.
- Engineered an active LC L-match output network and multi-finger gate topologies (30 fingers of $5.0\ \mu\text{m}$) to neutralize the Miller effect, ensuring precision $50\ \Omega$ impedance matching ($S_{11} = -17.06\ \text{dB}$, $S_{22} = -23.27\ \text{dB}$).
- Verified high power efficiency via ADE-L DC operating point extraction, yielding a low power consumption (P_{DC}) of 14.13 mW and an exceptional Figure of Merit (FoM) of 21.05 GHz/mW for X-band receiver front-ends.
- Actively collaborated with the research group, presenting weekly simulation data, design trade-offs, and performance metrics to the principal investigator to iteratively optimize the RF front-end architecture.

ENGINEERING PROJECTS

Dual-Input LiPo Battery Charger PCB | Altium Designer

Aug 2026

- Designed a custom 2-layer PCB for a lithium-ion battery charger, including dual power inputs (Micro-USB and DC barrel jack) safely separated by input OR-ing diodes.
- Designed the layout with thick power traces for safe battery charging and used a bottom copper pour to connect all grounds and keep the board cool.
- Added a physical solder jumper to let users switch between 100mA and 500mA charging speeds, mapped 3D component bodies to check clearances, and printed clear silkscreen safety warnings.

Automated CMOS Inverter Characterization | Python, NGSPICE, Pandas

July 2026 – Aug 2026

- Wrote a Python automation script that runs NGSPICE silently in the background, automatically sweeping 180nm CMOS transistor widths from $1.0\ \mu\text{m}$ to $5.0\ \mu\text{m}$. Used Pandas to extract and parse the raw simulation data directly from text files, replacing manual GUI testing to efficiently analyze Voltage Transfer Characteristics (VTC).
- Graphed the shifting switching thresholds using Matplotlib to visually prove how changing the PMOS size alters the strength of the pull-up network.

180nm 6T SRAM Cell Design and Characterization | Analog VLSI Design

July 2026

- Designed a 6T SRAM cell in a 180nm CMOS process using Cadence Virtuoso, optimizing transistor sizing (Cell Ratio = 2.0, Pull-Up Ratio = 0.7) for balanced read/write stability.
- Extracted Voltage Transfer Characteristics (VTC) and applied Seevinck's method via a custom MATLAB script to calculate a robust Read Static Noise Margin (SNM) of 309.0 mV.
- Performed transient and DC analyses to validate energy efficiency, achieving an active dynamic power of 86.39 μ W (at a 40ns period) and an ultra-low static leakage of 24.1 pW.

CERTIFICATIONS

Microsensors and Nanosensors (In Progress) | NPTEL / IIT Guwahati

July 2026 – Oct 2026

System Design Through Verilog (In Progress) | NPTEL / IIT Guwahati

July 2026 – Sept 2026

VLSI for Beginners | NIELIT Calicut (MeitY C2S Project)

Aug 2026

Introduction on Intellectual Property to Engineers | NPTEL / IIT Kharagpur

Jan 2026 – Mar 2026

WORKSHOPS & INDEPENDENT STUDY

Radio-Frequency Integrated Circuits (Independent Audit) | NPTEL / IIT Kanpur

July 2026 – Present

Seminar: VLSI Testing and Physical Design | MITS ISTE Student Chapter

Feb 2026

Seminar: VLSI and Electromagnetic Industry Practices | MITS IETE Students Forum

Feb 2026

Guest Talk: India's Semiconductor Revolution & VLSI | MITS Alumni Association

Aug 2025

HONORS & AWARDS

1st Prize (INR 25,000), Innovation Nexus 2026 Hackathon | MITS, PathVision & IITNiF

Aug 2026

- Awarded 1st place out of 50 competing teams during a rigorous 16-hour hackathon for solving a targeted industry problem statement to develop a "Real-Time Agricultural Transport Condition Monitoring System".
- Collaborated with my team to design the hardware architecture and IoT data ingestion pipeline using an ESP32 and multi-modal sensors (DHT11, MQ-4, GPS).

National Level Technical Symposium Wins | Multiple Institutions

2024 – 2026

- 2nd Position, "Circuit Detective":** ECLECTICA 2K26 at Madanapalle Institute of Technology and Science (April 2026). Awarded a cash prize.
- Double 1st Prize, "Circuit Hunt" & "Technical Quiz":** E-Merge 2K24 at JNTUACEA (Nov 2024). Awarded a total of INR 2000 cash prize for winning both events.
- 1st Prize, Paper Presentation:** Ripple National Symposium at RGM College of Engineering, Nandyal. Awarded an INR 1000 cash prize.

Consistent Academic Excellence & Campus Awards | ESC Govt Polytechnic College

2023 – 2025

- 5th Sem:** Awarded a wooden memento and merit certificate on College Annual Day (May 2025) for top examination results.
- 4th Sem:** Awarded the "Optical and Mobile Communication" textbook by faculty for academic performance (Nov 2024).
- 3rd Sem:** Received an INR 2000 cash prize, a "French Crown" pen, and the "Integrated Electronics" textbook (McGraw Hill) from faculty for outperforming in academics.
- Campus Quiz Winner:** Secured 1st place in the Independence Day Quiz (August 2023) and 2nd place in the Republic Day Quiz (January 2023), receiving mementos for both.